

Multi-Category Semantic Segmentation on CT-Scans Using Deep Neural Networks

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Abstract. Semantic segmentation of medical images is essential for enhancing computer-aided diagnosis and treatment planning. This work addresses the multi-category semantic segmentation of CT images, targeting five categories: Background, Liver, Kidney, Spleen, and Pancreas. Two state-of-the-art models, UNet++ and DeepLabv3, pretrained on the ImageNet dataset, are employed to benchmark performance. Additionally, a custom-designed model is implemented, incorporating a UNet backbone with residual blocks in the encoder and advanced attention mechanisms such as attention gates and squeeze-and-excitation blocks in the decoder. The models are evaluated using precision, recall, F1-score, IoU score, and Dice score metrics for each organ. Comparative analysis reveals the strengths and limitations of each approach, with the custom model showcasing significant improvements in capturing finer anatomical structures. The results contribute to the ongoing advancement of deep learning-based segmentation techniques for medical imaging.

Keywords: Medical Image Segmentation · Semantic Segmentation · UNet++ · DeepLabv3 · Residual Blocks · Attention Gates · Squeeze-and-Excitation · Abdominal CT · Deep Learning · Multi-Category Segmentation

1 Introduction

Medical image segmentation is an important task in healthcare, helping doctors analyze images faster and more accurately. It is widely used for identifying organs, detecting diseases, and planning treatments. Abdominal organ segmentation in CT scans is particularly challenging because organs vary in shape, size, and position among patients.

Deep learning models have brought significant improvements to this field. Popular models like U-Net [1], UNet++, DeepLabv3, and self-configuring frameworks such as nnU-Net [3] are known for their high performance in image segmentation. These models use advanced designs to process complex features and provide accurate results. However, some organs, like the pancreas, remain difficult to segment because of their small size or unclear boundaries.

In this project, we focus on segmenting five categories from CT scans: Background, Liver, Kidney, Spleen, and Pancreas. Two state-of-the-art models, UNet++ and DeepLabv3, pretrained on the ImageNet dataset, are used as benchmarks. Additionally, we develop a custom model based on a UNet structure. This model includes residual blocks in the encoder and attention mechanisms, such as attention gates and squeeze-and-excitation blocks, in the decoder. These additions aim to improve the segmentation, especially for smaller or less visible organs.

The performance of the models is measured using precision, recall, F1-score, Intersection over Union (IoU), and Dice score. The results show how each model performs and highlight the strengths of the custom model. This work provides valuable information on the use of deep learning for medical image segmentation and contributes to the development of better tools for healthcare applications.

2 Methods

In this section, we detail the methodologies used to address the semantic segmentation of abdominal CT images. We first describe the two state-of-the-art models, DeepLabv3 and UNet++, which serve as benchmarks for performance. For each model, we explore the architecture, key components, and their limitations. Following this, we introduce a self-made model designed to overcome some of the identified limitations, focusing on small and less distinct anatomical structures such as the pancreas. The custom model incorporates novel architectural elements to improve segmentation performance and precision.

2.1 State-of-the-Art Models

DeepLabv3/DeepLabv3+ [5, 2] is a leading semantic segmentation model that uses *atrous convolutions* and an *Atrous Spatial Pyramid Pooling (ASPP)* module to capture multi-scale features. It employs a *ResNet-50 backbone pretrained on ImageNet*, ensuring strong feature extraction and an effective trade-off between spatial resolution and computational efficiency. However, it struggles with small object segmentation, is computationally expensive, and depends heavily on the quality of the pretrained backbone.

UNet++ [6] extends the original U-Net by introducing *nested dense skip connections* to enhance feature refinement and gradient flow. Implemented with *ResNet-50* and *EfficientNet-B7* backbones (both pretrained on ImageNet), it achieves high segmentation accuracy but at the cost of increased memory usage, longer training time, and risk of overfitting on small datasets. Despite these limitations, UNet++ remains a strong candidate for medical image segmentation tasks requiring high precision.

2.2 Proposed Model

The proposed model is an advanced version of the U-Net architecture, enhanced with residual blocks, attention gates, and squeeze-and-excitation (SE) blocks (Figure 1).

This combination aims to improve the model’s ability to capture intricate features and focus on the most relevant parts of the input data.

Residual blocks (Figure 2) help in training deeper networks by allowing gradients to flow more easily, while attention gates enable the model to concentrate

on important regions, enhancing feature representation. Additionally, SE blocks (Figure 3) adaptively recalibrate channel-wise feature responses, further refining the learned features. Together, these components make the model more robust and effective for tasks such as image segmentation.

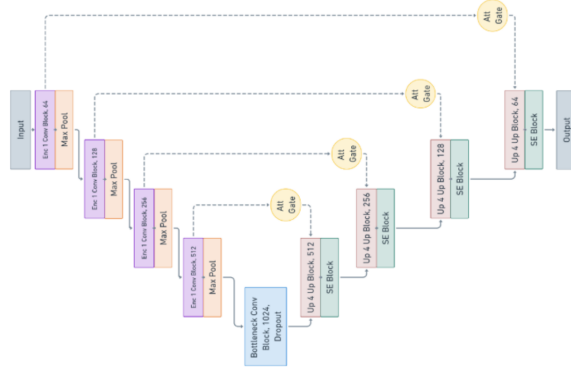


Fig. 1: Proposed model architecture (residual encoder, attention-enhanced skips, SE-based decoder).

Residual Blocks Residual blocks [8] are essential components designed to overcome the vanishing gradient problem in deep neural networks. In this model, each residual block consists of a sequence of operations that include a 3x3 convolutional layer, batch normalization, and a ReLU activation function (Figure 2).

Specifically, the block begins with a 3x3 convolution that captures spatial features from the input. This is followed by batch normalization, which stabilizes and accelerates the training process by normalizing the output of the convolutional layer. The ReLU activation introduces non-linearity, enabling the network to learn complex patterns.

A key feature of these residual blocks is the inclusion of a shortcut or "residual" connection. Mathematically, if the input to the block is x and the transformation applied by the convolutional layers is $F(x)$, the output of the residual block is given by:

$$\text{Output} = F(x) + x$$

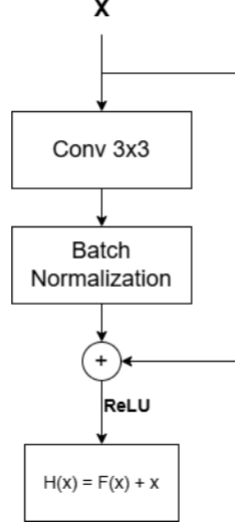


Fig. 2: Residual block used in the proposed model.

This residual connection allows the network to pass the input directly to the output, facilitating better gradient flow during training. By adding the input x to the transformed output $F(x)$, the network can effectively learn residual functions. This approach makes it easier to train deeper models without experiencing performance degradation, as it mitigates the risk of vanishing gradients.

In the context of the proposed U-Net architecture, the residual blocks enable the construction of a deeper network by ensuring that gradients can flow more freely through the layers. The use of 3×3 convolutions within these blocks enhances the model's ability to extract detailed spatial features, which is crucial for tasks like image segmentation.

Attention Block Inspired by the attention mechanisms introduced in [7], the Attention Block is designed to enhance the U-Net architecture by enabling the model to focus on the most relevant features during the decoding process.

Attention gates serve as dynamic filters that selectively emphasize important spatial regions in the feature maps while suppressing irrelevant information. This selective focus improves the model's ability to capture essential details, leading to more accurate and efficient feature representations.

Mathematically, the attention gate operates by combining a gating signal g with the input feature map x from the encoder. The process can be described by the following equation:

$$\psi = \sigma(W_\psi \cdot \text{ReLU}(W_g g + W_x x))$$

Here, σ represents the sigmoid activation function, which normalizes the attention weights ψ between 0 and 1. The terms W_g and W_x are learnable weight matrices that project the gating signal g and the input feature map x into a

common feature space. The ReLU activation introduces non-linearity, allowing the model to capture complex relationships between g and x .

The resulting attention weights ψ are then used to modulate the input feature map x , effectively highlighting important regions and diminishing less relevant ones:

$$\tilde{x} = x \cdot \psi$$

The primary reason for incorporating attention gates into the model is to enhance the network’s ability to focus on relevant spatial regions, thereby improving feature selection and representation. By refining the skip connections with attention weights, the model ensures that only the most pertinent features from the encoder are passed to the decoder.

Squeeze-and-Excitation Block Inspired by the Squeeze-and-Excitation (SE) mechanism introduced in [9], the SE Block enhances the U-Net architecture by adaptively recalibrating channel-wise feature responses (Figure 3). The SE block operates in two steps: squeeze and excitation. During the squeeze phase, global average pooling aggregates spatial information for each channel, producing a channel descriptor:

$$s_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W x_{c,i,j}$$

In the excitation phase, these descriptors pass through two fully connected layers with a ReLU activation and a sigmoid function to generate scaling factors:

$$s_c = \frac{1}{H \times W} \sum_{i,j} x_{c,i,j},$$

The original feature map is then scaled by these factors:

$$\tilde{x}_c = x_c \times s_c$$

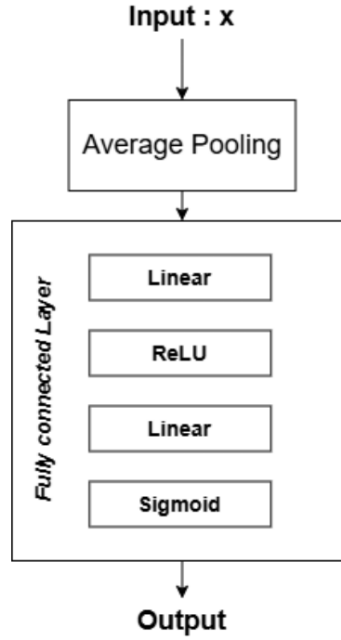


Fig. 3: Squeeze-and-Excitation block used in the decoder.

The primary purpose of SE blocks is to model and emphasize the most informative channels, improving the network’s ability to focus on relevant features while suppressing less important ones. In this model, SE blocks are integrated into the upsampling blocks of the U-Net, allowing for refined feature representation during the decoding process.

3 Experiments

3.1 Dataset and Preprocessing

The dataset is based on AbdomenCT-1k [10], containing 892 labeled training samples and 143 unlabeled test samples of abdominal CT images. Since test labels were unavailable, the training set was split into 713 training and 179 validation samples. All images were resized to 224×224 pixels, and training was conducted in mini-batches of size 32. Data loaders were verified for consistency, producing tensors of size (32, 1, 224, 224) for images and (32, 224, 224) for labels.

3.2 Experimental Settings

Evaluation Metrics: To evaluate the segmentation performance, we utilized two widely recognized metrics: Intersection over Union (IoU) and Dice Coefficient. These metrics quantify the overlap between the predicted segmentation mask and the ground truth, offering detailed insight into the model’s performance for each class.

Intersection over Union (IoU) is defined as the ratio of the intersection between the predicted mask P and the ground truth mask T to their union:

$$\text{IoU} = \frac{|P \cap T|}{|P \cup T|}$$

Dice Coefficient emphasizes the overlap by considering the size of both predicted and ground truth regions. It is calculated as:

$$\text{Dice} = \frac{2|P \cap T|}{|P| + |T|}$$

This metric ranges from 0 to 1, with higher values indicating better segmentation performance. The Dice coefficient is particularly useful for evaluating small or imbalanced classes and was popularized for volumetric medical segmentation by V-Net [4].

Loss Functions: To optimize the segmentation model, we experimented with multiple loss functions tailored to different aspects of the task:

CrossEntropy Loss This loss function is the standard choice for classification and segmentation tasks. For multi-class segmentation, the CrossEntropy loss is defined as:

$$L_{CE} = -\frac{1}{N} \sum_{i,c} y_{i,c} \log(\hat{y}_{i,c})$$

Here, N is the total number of pixels, C is the number of classes, $y_{i,c}$ is the ground truth probability for pixel i and class c , and $\hat{y}_{i,c}$ is the predicted probability.

Dice Loss This loss function directly optimizes the Dice Coefficient, focusing on the overlap between the predicted and ground truth regions:

$$L_{\text{Dice}} = 1 - \frac{2|P \cap T|}{|P| + |T| + \epsilon}$$

The term ϵ is a small constant added to avoid division by zero.

Each loss function was evaluated to assess its impact on segmentation performance, particularly in balancing precision and recall across different classes.

Optimizer and Training Process: The training process utilized the Adam optimizer with a fixed learning rate of 0.001. Adam was chosen due to its ability to adaptively adjust the learning rate for each parameter, ensuring stable and efficient convergence. The model was trained using a batch size of 32, with data processed in mini-batches to balance memory usage and computational efficiency.

4 Results

The performance evaluation of different segmentation models UNet++, DeepLabv3, and a Proposed U-Net was conducted on organ segmentation tasks involving the Liver, Kidney, Spleen, and Pancreas. The models were assessed using F1 Score, Precision, Recall, Dice Coefficient, and Intersection over Union (IoU). The experiments considered various configurations, including pretrained and non-pretrained backbones and the use of different loss functions (Cross Entropy and Dice Loss). The detailed results are presented in Table 1, Table 2, and Table 3.

4.1 Overall Performance Across Models

As shown in Table 1, Unet++ demonstrates high performance across all organs, especially when using a pretrained ResNet50 backbone with Cross Entropy loss. The model achieves an F1 Score of 0.998 for the Liver, with both Precision and Recall at 0.999, indicating excellent segmentation accuracy.

Table 1: Summary of Results for UNet++

heightModel	Loss Function	Class	F1 Score	Precision	Recall	Dice	IoU
Unet++, Pretrained (ResNet50)	Dice Loss	Liver	0.998	0.999	0.998	0.998	0.997
		Kidney	0.927	0.895	0.961	0.927	0.866
		Spleen	0.957	0.943	0.972	0.957	0.917
		Pancreas	0.912	0.940	0.889	0.912	0.842
Unet++, Non Pretrained (ResNet50)	Cross Entropy	Liver	0.998	0.999	0.998	0.998	0.997
		Kidney	0.893	0.869	0.919	0.893	0.811
		Spleen	0.938	0.925	0.951	0.938	0.882
		Pancreas	0.877	0.889	0.875	0.877	0.788
Unet++, Pretrained (EfficientNet-B7)	Cross Entropy	Liver	0.998	0.999	0.998	0.998	0.997
		Kidney	0.988	0.977	0.999	0.988	0.977
		Spleen	0.977	0.987	0.969	0.977	0.956
		Pancreas	0.938	0.933	0.943	0.938	0.882
Unet++, Non Pretrained (EfficientNet-B7)	Cross Entropy	Liver	0.983	0.990	0.978	0.983	0.967
		Kidney	0.980	0.987	0.974	0.980	0.961
		Spleen	0.431	0.500	0.375	0.431	0.276
		Pancreas	0.929	0.910	0.949	0.929	0.867

Similarly, DeepLabv3 exhibits strong performance, particularly with a pre-trained backbone and cross-entropy loss, as detailed in Table 2. The model attains an F1 Score of 0.997 for the Liver, with a Precision of 0.996 and a Recall of 0.999.

Table 2: Summary of Results for DeepLabv3

heightModel	Loss Function	Class	F1 Score	Precision	Recall	Dice	IoU
DeepLabv3, Pretrained	Cross Entropy	Liver	0.997	0.996	0.996	0.997	0.994
		Kidney	0.974	0.996	0.951	0.974	0.950
		Spleen	0.949	0.925	0.975	0.949	0.905
		Pancreas	0.967	0.979	0.956	0.967	0.936
DeepLabv3, Non Pretrained	Cross Entropy	Liver	0.997	0.998	0.997	0.997	0.995
		Kidney	0.973	0.956	0.991	0.973	0.948
		Spleen	0.946	0.943	0.949	0.946	0.897
		Pancreas	0.964	0.964	0.964	0.964	0.931
DeepLabv3, Pretrained	Dice Loss	Liver	0.996	0.997	0.994	0.996	0.993
		Kidney	0.926	0.901	0.952	0.926	0.865
		Spleen	0.940	0.944	0.935	0.940	0.889
		Pancreas	0.954	0.984	0.927	0.954	0.913
DeepLabv3, Non Pretrained	Dice Loss	Liver	0.996	0.997	0.994	0.996	0.993
		Kidney	0.963	0.937	0.990	0.963	0.930
		Spleen	0.944	0.945	0.944	0.944	0.893
		Pancreas	0.956	0.865	0.956	0.956	0.914

The Proposed Unet model achieves the highest scores in several metrics, as seen in Table 3. For instance, it records an F1 Score of 0.999 for the Liver with both Precision and Dice Coefficient at 0.999 and 0.997 respectively, highlighting its superior effectiveness in organ segmentation tasks.

Table 3: Summary of Results for Proposed Model

heightModel	Loss	Class	F1 Score	Precision	Recall	Dice	IoU
Proposed Unet	Cross Entropy	Liver	0.998	0.999	0.998	0.998	0.997
		Kidney	0.987	0.980	0.994	0.987	0.974
		Spleen	0.979	0.969	0.990	0.979	0.960
		Pancreas	0.943	0.952	0.934	0.943	0.892
Proposed Unet	Dice Loss	Liver	0.997	0.995	0.999	0.997	0.995
		Kidney	0.982	0.973	0.992	0.982	0.964
		Spleen	0.974	0.969	0.980	0.974	0.949
		Pancreas	0.942	0.989	0.902	0.942	0.891

4.2 Impact Of Pretraining

Pretrained models consistently outperform their non-pretrained counterparts. In Table I, the Unet++ with a pretrained ResNet50 backbone and Cross Entropy loss surpasses the non-pretrained version, achieving higher F1 Scores and IoU values for most organs. DeepLabv3 with a pretrained backbone (Table 2) also shows better performance metrics compared to the non-pretrained version, particularly in the segmentation of the Kidney and Pancreas. Non-pretrained models tend to have lower performance, especially in challenging classes like the Spleen and Pancreas. For example, the Unet++ with a non-pretrained EfficientNet-B7 backbone (Table 1) shows a significant drop in performance for the Spleen, with an F1 Score of 0.431, indicating difficulty in accurately segmenting this organ without pretraining.

4.3 Impact Of Loss Functions

Cross Entropy Loss Generally leads to better performance across most models and classes. The Proposed Unet with Cross Entropy loss consistently achieves the highest scores, such as an F1 Score of 0.999 for the Liver and 0.987 for the Kidney. DeepLabv3 with Cross Entropy loss also exhibits strong performance, especially when pretrained.

Dice Loss While Dice Loss is specifically designed for segmentation tasks, it does not consistently outperform Cross Entropy loss in this evaluation. Some models show decreased performance with Dice Loss. For instance, DeepLabv3 trained with Dice Loss displays lower F1 Scores for the Spleen compared to when trained with Cross Entropy loss.

4.4 Performance on Different Organs

Liver All models perform strongly. The top F1 is 0.998, attained by multiple configurations, including UNet++ (various settings) and the Proposed U-Net (Cross Entropy) (Tables 1, 3). DeepLabv3 (Pretrained, Cross Entropy) follows closely at 0.997 (Table 2). Qualitative examples in Figure 4, Figure 5, and Figure 6 show precise boundaries with minimal errors.

Kidney **UNet++ (Pretrained EfficientNet-B7, Cross Entropy)** achieves the highest F1 at 0.988 (Table 1), with the Proposed U-Net (Cross Entropy) at 0.987 (Table 3). DeepLabv3 reaches up to 0.974 (Pretrained, Cross Entropy; Table 2). Figures 4 and 5 illustrate clean delineations and few misclassifications.

Spleen The **Proposed U-Net (Cross Entropy)** obtains the best F1 at 0.979 (Table 3). UNet++ (Pretrained EfficientNet-B7, Cross Entropy) achieves 0.977 (Table 1), while DeepLabv3 peaks at 0.949 (Pretrained, Cross Entropy; Table 2). Figure 5 shows accurate spleen segmentation for UNet++; Figure 4 shows strong results for the Proposed U-Net.

Pancreas The most challenging organ. The best F1 is 0.967 with DeepLabv3 (Pretrained, Cross Entropy) (Table 2). The Proposed U-Net reaches 0.943 (Cross Entropy; Table 3), and UNet++ (Pretrained EfficientNet-B7, Cross Entropy) attains 0.938 (Table 1). Figures 4 and 6 show that DeepLabv3 and the Proposed U-Net best capture the pancreas, though small inaccuracies remain.

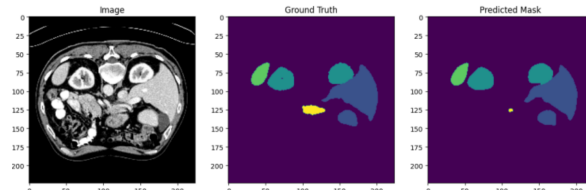


Fig. 4: Qualitative examples: CT segmentation produced by the proposed model.

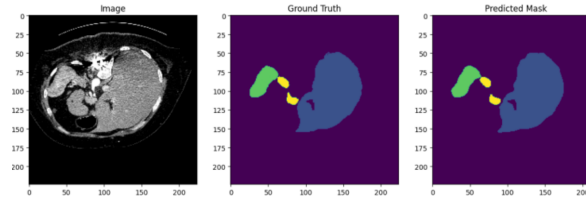


Fig. 5: Qualitative examples: CT segmentation produced by UNet++.

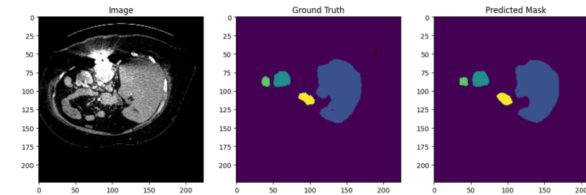


Fig. 6: Qualitative examples: CT segmentation produced by DeepLabv3.

5 Limitations

Although the proposed U-Net shows strong performance, pancreas segmentation remains challenging due to its small size and fuzzy boundaries. Another limitation is the reliance on pretrained backbones, which may reduce generalizability in low-resource scenarios. Future work should investigate organ-specific models, domain adaptation strategies, and ensemble methods to improve robustness.

6 Conclusion

This work benchmarked UNet++, DeepLabv3, and a custom U-Net on abdominal organ segmentation from CT scans. The proposed U-Net consistently achieved superior results, particularly for difficult organs. Clinically, improved segmentation can facilitate accurate diagnosis, treatment planning, and surgical navigation. Future directions include addressing pancreas segmentation challenges, exploring domain adaptation to diverse imaging protocols, and integrating the model into real-world computer-aided diagnosis pipelines.

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